

**Artificial Intelligence
CS-451**

PROJECT PROPOSAL



Instructor

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AI-Based Handwriting Analysis for Neurological Detection (NeuroScript)

1. Problem Statement (Innovation Focus)

Real-World Problem

Neurological and behavioral disorders such as Attention Deficit Hyperactivity Disorder (ADHD) and Obsessive-Compulsive Disorder (OCD) are often diagnosed late due to dependence on traditional clinical methods. Current diagnostic practices mainly rely on:

- Behavioral observation
- Structured clinical interviews
- Subjective assessments by practitioners

These methods are often:

- Time-consuming
- Practitioner-dependent
- Inconsistent across evaluations
- Difficult to scale for continuous monitoring

Additionally, there is a lack of localized Pakistani datasets and very limited use of real-time handwriting behavior in neurological screening systems.

Why It Is Important

Early and accessible screening can:

- Improve treatment outcomes through timely intervention
- Reduce long-term cognitive and behavioral complications
- Enable continuous and non-invasive monitoring
- Support clinicians with data-driven insights
- Increase accessibility of neurological screening in educational and healthcare environments

Novelty of the Idea

The proposed system, **NeuroScript**, introduces an AI-powered handwriting analysis framework that combines:

- Real-time handwriting input analysis using stylus/finger
- Detection of neuromotor patterns such as:
 - Spacing irregularities
 - Letter formation inconsistencies
 - Stroke variations
 - Writing speed and movement patterns
- Explainable AI (XAI) for transparent predictions

- Human-readable AI-generated explanations for clinicians and users

Unlike traditional image-only systems, NeuroScript leverages sequential handwriting behavior to capture dynamic neuromotor signals associated with ADHD and OCD.

Target Users / Stakeholders

- Clinicians and psychologists
- Educational institutions and counselors
- Neuroscience and AI researchers
- Students and individuals seeking early screening
- Healthcare technology organizations

2. Literature Gap / Motivation

Existing Approaches

Traditional neurological diagnosis methods primarily depend on:

- Clinical interviews
- Behavioral observations
- Standardized psychological tests

Although some AI-based systems exist, most:

- Focus only on classification accuracy
- Lack interpretability
- Ignore real-time handwriting behavior
- Use static image-based analysis only

Limitations

Current systems suffer from:

- High subjectivity in diagnosis
- Lack of continuous monitoring
- Limited use of behavioral biometrics
- Poor clinician trust due to black-box AI predictions
- Absence of localized Pakistani handwriting datasets

Research Gap

There is currently no widely adopted system that:

- Uses handwriting as a primary neurological signal
- Captures real-time handwriting behavior

- Provides interpretable AI-based predictions
- Combines predictive AI with clinician-friendly explanations
- Utilizes culturally relevant Pakistani handwriting data

Motivation

The motivation behind NeuroScript is to:

- Enable early, non-invasive neurological screening
- Utilize handwriting as an underexplored behavioral biomarker
- Improve trust through Explainable AI techniques
- Provide accessible and scalable screening support
- Develop a localized AI solution suitable for Pakistani users

3. Dataset Details

Dataset Source

The system will use a Pakistani handwriting dataset consisting of:

- Handwriting samples collected from Pakistani students
- English handwriting samples
- Stylus/finger-based digital handwriting input
- Paper-based handwritten samples converted into digital form

Potential supplementary datasets may also be collected from public research repositories if needed.

Type of Data

The dataset will include:

- Sequential handwriting stroke data
- Stylus/finger movement coordinates
- Timing and speed information
- Digitized handwriting images

Number of Samples

Approximately:

- 500–5,000 handwriting samples

The dataset will cover multiple age groups to improve generalization.

Features

The system will analyze:

- Stroke order
- Writing speed
- Pen movement patterns
- Letter formation
- Word and letter spacing
- Alignment consistency
- Stroke variation
- Temporal handwriting behavior

Target Variable

Classification labels include:

- ADHD Indicators
- OCD Indicators
- Normal / No Disorder

Data Preprocessing Plan

Preprocessing steps include:

- Noise removal and image cleaning
- Stroke normalization
- Segmentation of handwriting sequences
- Feature extraction from temporal handwriting signals
- Coordinate scaling and normalization
- Sequence padding for LSTM compatibility

Synthetic Data Generation

To improve model robustness and dataset balance:

- Data augmentation techniques may be applied
- Synthetic sequence variations may be generated

Techniques include:

- Rotation
- Scaling
- Temporal variation simulation
- Sequence perturbation

4. Proposed Predictive AI Model

Problem Type

- Multiclass Classification Problem

Long Short-Term Memory (LSTM)

The primary model used in NeuroScript will be a Long Short-Term Memory (LSTM) network.

LSTM is a type of recurrent neural network (RNN) specifically designed for sequential and time-dependent data. It is highly suitable for analyzing handwriting motion and neuromotor behavior.

The model will learn patterns from:

- Stroke order
- Writing speed
- Pen movement
- Timing behavior
- Sequential handwriting dynamics

Unlike static image-based approaches, LSTM captures real-time behavioral patterns, making predictions more meaningful and neurologically relevant.

Why LSTM Is Suitable

- Handles sequential handwriting data effectively
- Captures temporal dependencies
- Learns dynamic neuromotor behavior
- Suitable for stylus/finger handwriting analysis
- Improves behavioral pattern recognition accuracy

Evaluation Metrics

The system will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

5. Proposed XAI Techniques

Techniques Used

The system will integrate:

- SHAP (SHapley Additive exPlanations)
- LIME (Local Interpretable Model-Agnostic Explanations)

Why These Techniques Are Suitable

SHAP

- Provides global and local feature importance
- Quantifies feature contributions
- Helps identify influential handwriting traits

LIME

- Explains individual predictions
- Supports case-by-case analysis
- Improves transparency for clinicians

What the System Will Explain

The XAI layer will highlight:

- Irregular spacing
- Stroke variation
- Inconsistent letter formation
- Writing speed anomalies
- Pen movement behavior

Presentation of Explanations

Explanations will be displayed through:

- Feature importance visualizations
- Highlighted handwriting regions
- Interactive dashboard insights
- Text-based interpretation summaries

This improves clinician trust and supports decision-making.

6. Proposed Generative AI Component

Approach

The system will integrate a Generative AI component using:

- OpenAI API
- or
- Open-source Large Language Models (LLMs)

Integration with Predictive System

Pipeline flow:

- Handwriting input
 - Preprocessing
 - LSTM prediction
 - XAI explanation
 - Generative AI interpretation
 - Final diagnostic report

Value Addition

The Generative AI layer converts technical model outputs into:

- Human-readable explanations
- Simplified clinical summaries
- Suggested next steps
- Easy-to-understand recommendations

Key Outputs

The system will generate:

- Diagnostic insight reports
- Plain-language explanations
- Behavioral observations
- Recommendations for professional evaluation

7. System Architecture

Pipeline Flow

1. Handwriting Input (Stylus/Finger or Digitized Samples)
2. Preprocessing & Feature Extraction
3. Sequential Data Preparation
4. LSTM-Based Prediction Model
5. Explainable AI Layer (SHAP + LIME)
6. Generative AI Interpretation Layer
7. Interactive Web Dashboard Output

8. Tools & Technologies

Programming Language

- Python

Libraries & Frameworks

Deep Learning

- TensorFlow

Data Processing

- NumPy
- Pandas

Explainable AI

- SHAP
- LIME

Image & Sequence Processing

- OpenCV

Generative AI

- OpenAI API / HuggingFace Transformers

Front-End / Deployment

- Streamlit

9. Implementation Requirements

Development Requirements

- Entire system developed in Python
- Modular and maintainable codebase
- Proper documentation for each module
- Version control using GitHub

User Interface

The system will provide an interactive web-based interface using:

- Streamlit

System Features

- Real-time handwriting input using stylus/finger
- Upload handwriting samples
- AI-based neurological prediction
- Explainable AI visualizations
- Human-readable diagnostic summaries
- Interactive dashboard for analysis and reporting